LESSON PLAN · SCRATCH (GR 3–5) INTERMEDIATE: STORY, QUIZ & MUSIC STUDIO

Lesson 3: Project: Pattern Art Gallery

Module 1 | 50 minutes | Project: pattern art | Grades 3–5 · ages 8–10

Objective: Plan, build and share an art gallery of at least two generated patterns using nested loops and variables.

Success criteria

- My gallery has two different click patterns.
- One uses a nested loop and one a growing variable.
- I can explain how changing a number changes the art.

Key vocabulary

- **generative art**: Art made by a set of rules (code).
- **parameter**: A number you can change to alter the result.
- **gallery**: A collection of works to show visitors.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Human spirograph (5 min): a student holds a rope end in the middle, another walks the pattern; the class gives the rule (“step, turn”) and changes one number to see a new shape.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
25 min	**Build**	Build-along steps, then Which idea makes each pattern?. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other’s work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; Project: Pattern Art Gallery.

Unplugged warm-up

Human spirograph (5 min): a student holds a rope end in the middle, another walks the pattern; the class gives the rule (“step, turn”) and changes one number to see a new shape.

Teaching the idea

The project consolidates nested loops and variables-in-loops. Learners build two contrasting generators triggered by click events and reflect on how parameters change outcomes.

Analogy: A pattern generator is like a recipe for art: change one ingredient (a number) and you get a new dish.

Live demo: Show a finished gallery; live-change one number (the turn angle) and redraw. Ask learners to predict the effect first.

Questions to ask

Ask	Listen for
Which number has the biggest effect?	Usually the turn angle.
How do visitors choose a pattern?	They click an artist sprite.
What does your nested loop draw?	A shape repeated around a circle or in a row.

Differentiation

Support
Start from the two build-along scripts and change only colours and numbers.

Stretch
Add a third artist that uses BOTH ideas (a nested loop whose size grows), or keys to change colours.

Common misconceptions

Misconception	How to fix it
Each sprite shares one script	Scripts belong to a sprite; select each sprite to edit its code.
Erase all only clears my sprite's lines	It clears all pen drawing on the stage.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Nested loops	Missing	Inner loop works; outer repeat missing or misplaced	Correct nested loop draws a repeated shape	Nested loops used creatively (e.g. three levels, varied shapes)

Variables	No variable used	Variable made but not used in a block	Variable set, changed and used to grow a pattern	Variables control several parameters (size, angle, colour)
Events	Only the flag	One click works	Each artist responds to a click	Extra controls (keys) or a welcome screen
Design	Unplanned	Some colour choices	Attractive, clear gallery	Striking, intentional art
Reflection	Missing	Brief	Explains the effect of a change	Explains several experiments and choices

Exit ticket
One number I changed and what it did to my pattern.

Homework
Show your gallery at home and ask someone to choose their favourite. Note their reason.

Answer key — Kitabash activities

1. Which idea makes each pattern? (perform · categorize)

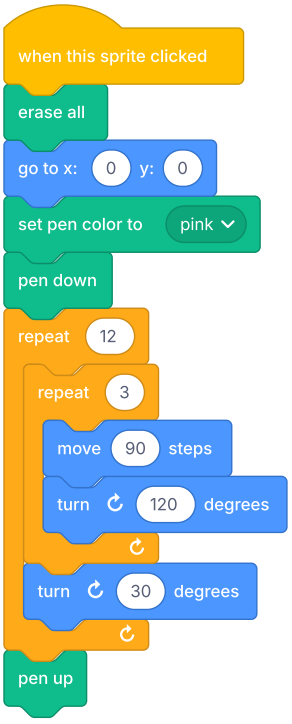
Nested loop: A flower of 12 triangles, A row of 5 squares
Growing variable: A spiral getting bigger, Lines that get longer each time

2. Project: Pattern Art Gallery (probe · code lab)

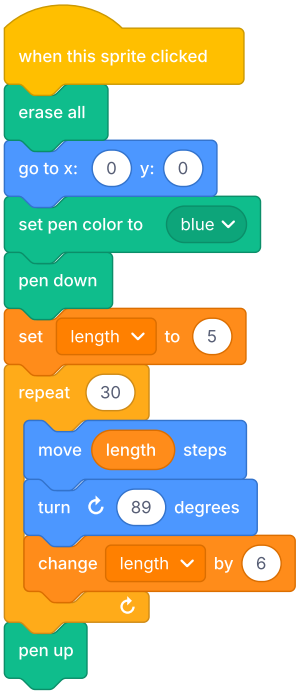
Goals checked: 2 or more artists · Clicking draws a pattern · A nested loop · A growing variable · Colours

Example project:

Pip



Butterfly



Answer key — worksheet

1. **Key words:** generative art = Art made by a set of rules (code). · parameter = A number you can change to alter the result. · gallery = A collection of works to show visitors.

2. **Which idea makes each pattern?:** Nested loop: A flower of 12 triangles, A row of 5 squares · Growing variable: A spiral getting bigger, Lines that get longer each time
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.